BUSINESS REQUIREMENT DOCUMENT

Riverside Power Distribution Company

1. DOCUMENT CONTROL:

Item	Detail
Sponsor	Managing Director (Mr. Tamuno Briggs)
Approver	Board
Business owner	Head of Network Operations
Change control	Any change to scope or targets requires sponsor approval

2. EXECUTIVE SUMMARY:

Outage complaints have risen to 118,000 per month, average outage duration to 11.2 hours, and only 54% of outages are restored within target against a 90% standard. NERC has issued a compliance notice, two industrial customers have moved to captive generation (₦2.4bn/yr revenue), and compensation claims are up 340%. The Board has asked for a recovery plan.

3. STAKEHOLDERS:

Stakeholder	Role	Interest and involvement
Mr. Tamuno Briggs	Managing Director	Sponsor; NERC relations; investor reporting
Mr. Emeka Eze	Head of Network Operations	Control centre, switching, restoration; process owner
Mr. Sunday Etim	Head of Field Technical	Repairs, maintenance, vegetation management; process owner
Mrs. Rukayat Bello	Head of Customer Service	Complaints and outage reporting; process owner
Mr. Chibueze Nwosu	Head of Metering	Meter and vending data; consulted
Mrs. Grace Oluwaseun	Head of Commercial	Revenue, industrial accounts; consulted
Mr. Ayodele Sanni	Chief Information Officer	SCADA, billing, metering and customer channels; delivery and data owner
Mr. Godwin Akpan	Chief Financial Officer	Cost control, investment appraisal, compensation; funding approver

Mrs. Halima Suleiman	Head of Regulatory	NERC reporting and compliance; owner of the regulator response
Shareholders	Core investor 60%, institutions 22%, state government 10%, employees 8%	Approval and engagement for network investment above ₦500m
NERC	Regulator	Compliance notice and restoration standard
Customers	Residential 596,000, Commercial 202,000, Industrial 42,000	Reliability and communication

4. BUSINESS PROBLEM AND OPPORTUNITY:

Problem: The Company cannot restore supply fast enough, cannot reliably detect where faults occur, and cannot prove which factors drive outages.

Opportunity: protect revenue from industrial and commercial customers (68% of revenue), avoid regulatory penalties, reduce compensation cost, and recover reputation.

5. SUMMARY OF THE PROBLEM:

Indicator	Then	Now
Complaints per month	42,000	118,000 (+181%)
Average outage duration	4.8 hrs	11.2 hrs (+133%)
Restoration within target	-	54% vs 90% target
Compensation claims	baseline	+340%
Industrial demand	-	2 customers self-generating (₦2.4bn/yr revenue lost)

6. EMPIRICAL OUTAGE DATA ANALYSIS (SUPPLEMENTARY):

The following findings are drawn from a supplementary statistical analysis of 10,000 cleaned historical outage log records covering RPDC's operating network (Aba, Calabar, Owerri, Port Harcourt, Uyo and Warri). They are presented here as corroborating evidence for the root-cause investigation; figures that could not be reconciled against the headline complaint and revenue figures elsewhere in this document have been excluded or flagged, consistent with BR-02/BR-03's requirement that investment be traceable to validated data.

Cause Code Category	Total Incidents	Incident Share	Mean MTTR (Hours)
Transformer Failure	1,699	16.99%	9.13 hrs
Cable Fault	1,684	16.84%	8.99 hrs
Unknown (Unmonitored)	1,677	16.77%	8.95 hrs

Overload Trip	1,672	16.72%	9.00 hrs
Protection Trip	1,661	16.61%	9.19 hrs
Vegetation Interference	1,607	16.07%	9.06 hrs
Total (Network-Wide)	10,000	100.0%	9.05 hrs

Importantly, transformer failure accounts for only 16.99% of recorded incidents directly challenging the assumption behind the originally proposed ₦6.8bn transformer-focused reinforcement programme, and reinforcing BR-02/BR-03's requirement that capital be allocated against diagnosed root causes rather than a single assumed cause.

Detection channel analysis across the same dataset shows SCADA (25.46%), the Network Operations Centre (25.08%), field crew discovery (24.85%) and customer calls (24.61%) each account for roughly a quarter of all fault detections an even split that is itself evidence of the telemetry gap: in a fully telemetered network, automated detection (SCADA) would be expected to dominate. This is consistent with the finding that 58 of the 96 substations have no telemetry (Section "CURRENT SYSTEM", SCADA row) and supports BR-05 and FR-09.

Dispatch delay analysis across all 10,000 outages shows a mean 210.7 minutes (3.51 hours) elapsed between fault detection and crew dispatch accounting for roughly 39% of the 9.05-hour average outage duration in this dataset, and attributed to manual ticket re-entry between the Service Desk and Field Scheduler. This directly supports FR-04 (linking Service Desk tickets to Outage Management records) and FR-06 (dispatch-priority rules).

Operational Area	Total Outage Incidents	Mean MTTR (Hours)
Aba	1,613	9.02 hrs
Calabar	1,732	9.16 hrs
Owerri	1,710	9.04 hrs
Port Harcourt	1,627	8.94 hrs
Uyo	1,700	9.16 hrs
Warri	1,618	9.01 hrs

7. STAKEHOLDERS WORKSHOP AND SURVEY QUESTIONS:

A quantitative customer survey of 200 and qualitative stakeholder interviews (executive leaders, operational heads, and customer representatives, coded using NVivo) were conducted to corroborate the empirical findings above with direct stakeholder evidence.

7.1 Quantitative Customer Survey Synthesis (N=200):

Survey Dimension	Distribution & Breakdown	Operational Requirement Implication
Q1. Customer Category	Industrial: 67 | Residential: 85 | Registered 20 | Commercial: 28	High commercial/industrial weight drives acute

		revenue exposure (supports BR-08).
Q2. Outage Frequency	Rarely:63 | Occasionally:83 | Frequently:22 | Very Satisfied: 17	75.0% dissatisfied widespread discontent with grid stability (BO-2).
Q3. Outage Duration	Dissatisfied: 78 | Very Dissatisfied: 65 | Neutral: 17 | Satisfied: 23 | Very Satisfied: 17	71.5% dissatisfied directly correlates with 11.2h MTTR (BO-2).
Q4. Biggest Impact	Financial Loss: 69| Operational Disruption: 77 | Alt. Power Cost: 21 | Equipment Damage: 19 | Other: 14	Direct monetary damage is pushing customers toward captive generation (BR-08).
Q5. Response Satisfaction	Dissatisfied: 80| Very Dissatisfied: 70 | Neutral: 12 | Satisfied: 18 | Very Satisfied: 20	71.0% dissatisfied slow dispatch and repair execution (BR-06, FR-06).
Q6. Communication	Dissatisfied: 72| Very Dissatisfied: 74| Neutral: 20 | Satisfied: 17| Very Satisfied: 17	76.5% dissatisfied lack of proactive status notifications (BR-10, FR-08).
Q7. Repeat Reporting	Very Dissatisfied: 83 | Dissatisfied: 69 | Neutral: 17 | Satisfied: 16 | Very Satisfied: 15	76.0% dissatisfied siloed Service Desk creates repetitive ticket logging (FR-04).
Q8. Desired Priority	Fewer Outages: 60 | Faster Restoration: 68| Better Communication: 27| Better Fault Detection: 23 | Other :22	Demand centres on speed of fault clearing and prevention (BO-1, BO-2).
Q9. Updates on Power Outage	Not Important: 85 | Slightly Important: 64| Moderately Important: 28| Very Important: 23 |	23 customers considered it important to receive updates on power outage.
Q10. Important Change	More consistent 75 | Frustrating: 63| Timing & staff handling the issue: 13| Less Consistent: 13 | Small obvious weakness :12	75 customers would like the power supply to be consistent.

7.2 Stakeholder Elicitation: Qualitative Themes (NVivo-Coded):

Interviews with executive and operational leaders were transcribed and thematically coded. Representative user stories elicited from this process are summarised below by theme.

Theme	Role	User Story Elicited
Regulatory Recovery	MD / CEO	As the CEO, I want to satisfy NERC's restoration standard and lift the compliance notice, so that we avoid penalties and licence risk.

Commercial Revenue Protection	MD / CEO	As the CEO, I want to restore confidence among industrial customers and halt the shift to captive generation, recovering lost revenue.
Infrastructure Investment	MD / CEO	As the CEO, I want to rigorously test capital solutions against operational data, so capital is deployed where it has the most impact.
Operational Efficiency	Operations Lead	As the Operations Lead, I want to review vegetation clearance contracts and headcount, addressing the root causes of plant-related line faults.
Control Room Visibility	Operations Lead	As the Operations Lead, I want telemetry installed across remote substations, so control room operators stop operating without visibility.

7.3 Customer User Stories (Representative Sample):

Customer Segment	User Story
Industrial / Manufacturing	As the customer, I want power supply to be stable and predictable, so that I can prevent damage to sensitive equipment and avoid halting production.
Commercial / SME	As the customer, I want consistent electricity supply, so that I can avoid material waste, missed delivery timelines, and heavy diesel expenses.
Residential	As the customer, I want reliable electricity at home, so that I can avoid unexpected costs, delays, and stress during basic daily tasks.
All segments (reporting)	As the customer, I want customer service to log my details correctly the first time, so that I do not have to repeat information or wait unnecessarily.
All segments (communication)	As the customer, I want proactive communication and realistic restoration estimates, so that I can plan my business and daily life.

8. CURRENT SYSTEM:

System	Limitation	Implication for this programme
SCADA	Only 38 of the 96 substations are equipped with telemetry systems. Consequently, faults at the remaining 58 substations are typically identified through customer complaints rather than automated monitoring and detection mechanisms.	At these sites, outage start times are often understated because faults are not detected when they occur. Instead, detection is delayed until customers report the outage, resulting in inaccurate outage-duration records and potentially affecting response-time measurements.
Outage Management	Cause codes often "unknown"; planned and unplanned not always separated; restoration recorded when crew reports back	Cause analysis unreliable; duration may be overstated at the end
Service Desk	Tickets not linked to outage records	Complaints cannot be reconciled to outages automatically
Field Scheduler	Manual dispatch on judgement; preventive work competes with faults	Priority is inconsistent.
Billing System	Estimated billing disputes	Some complaints may not be outage-related
Meter Data Management	Manual reads; vending outages	Limited use for outage verification today

9. BUSINESS OBJECTIVES:

ID	Objective	Measure	Baseline	Proposed target
BO-1	Meet NERC restoration standard	Restoration within target	54%	90% by date set in NERC notice
BO-2	Reduce outage duration	Average duration	11.2 hrs	Return toward 4.8 hrs baseline (phased)
BO-3	Reduce complaints	Unique complaints per month	118,000 (raw)	Toward 42,000 baseline, measured on deduplicated basis
BO-4	Protect industrial revenue	Demand retained and returned	−2 customers, ₦2.4bn/yr	No further loss; agreed recovery plan per customer
BO-5	Reduce compensation exposure	Claims value and count	+340%	Reversal of trend; target TBC

BO-6	Base investment on evidence	Share of capital tied to root-cause findings	0%	100% of approved spend traceable to findings
BO-7	Reliable outage measurement	Records with defined start and end, share coded "unknown", reporting by feeder, cause and detection method	Area reporting only	Agreed definitions; "unknown" share below a threshold to be set with Network Operations

10. PROJECT SCOPE BOUNDARIES:

In Scope

- Substation telemetry expansion: retrofitting monitoring capability across the unmonitored distribution substations identified under BR-05 and FR-09.
- Enterprise software integration: linking SCADA, Outage Management, Service Desk and Field Scheduler so outage records, tickets and dispatch share a common view (FR-04, FR-06, FR-13).
- Automated customer notification at outage detection, dispatch and restoration (FR-08).
- Reinstatement of preventive vegetation management, prioritised by fault risk (BR-07, FR-10).
- Field force automation: mobile work-order capture with cause coding and completion sign-off (FR-02, FR-14).

Out of Scope (unless a validated business case later confirms otherwise)

- Bulk transmission grid infrastructure operated by the Transmission Company of Nigeria (TCN).
- Modifications to NERC-regulated multi-year tariff orders (MYTO) or billing rates.
- Blanket, network-wide transformer replacement as a single aggregated initiative - the Financial Deliberation section below requires each improvement option to be costed and evaluated independently rather than bundled.

11. REQUIREMENTS:

In this section priority is determined using the MoSCoW framework (Must, Should, Could), while functional requirements are written to be solution-agnostic.

11.1 Business Requirements

ID	Requirement	Priority	Objective
BR-01	The company must be able to demonstrate compliance with the NERC restoration standard and reply to the compliance notice within the slated deadline	Must	BO-1
BR-02	The company must identify root causes of increased outage duration and frequency, supported by data	Must	BO-6
BR-03	Investment decisions must be traceable to diagnosed root causes with a costed business case	Must	BO-6
BR-04	The company must reduce repeated outages on feeders with high-fault.	Must	BO-2, BO-3
BR-05	The company must improve detection speed for faults on substations without telemetry	Must	BO-2
BR-06	The company must reduce restoration time through improved dispatch, prioritization and resourcing	Must	BO-1, BO-2
BR-07	The company must restore a preventive vegetation management schedule prioritized by fault risk	Should	BO-2
BR-08	The company must implement a plan to retain and recover industrial customer demand	Must	BO-4
BR-09	The company must manage compensation claims consistently and verifiably	Should	BO-5
BR-10	The company must communicate outage status and expected restoration to customers	Should	BO-3
BR-11	Define and enforce outage start, end and cause coding, and report by feeder, cause, detection method and urban or rural area	Must	BO-7
BR-12	Maintain complete transformer maintenance records	Should	BO-6

11.2 Stakeholder Requirements

ID	Stakeholder	Role	Requirement	Priority
SR-01	Managing Director and Board		Monthly recovery dashboard: NERC measure, duration by urban and rural area, over-24-hr outages, unique outages, contacts per outage, claims, revenue	Must
SR-02	Mr. Emeka Eze	Network Operations	Ranked list of worst feeders with cause and action status	Must
SR-03	Control room		Single view of open faults, crew assignments and priority	Must
SR-04	Mr. Sunday Etim	Field Technical	Fault and preventive job records with cause, materials and times	Should
SR-05	Mrs. Rukayat Bello	Customer Service	Agents can see known outages and estimated restoration time at the time of contact	Must
SR-06	Mrs. Halima Suleiman	Regulatory	Auditable restoration data in the form NERC requires	Must
SR-07	Mr. Godwin Akpan	Financial Department	Cost, benefit and payback for each intervention; verified compensation	Must
SR-08	Mrs. Grace Oluwaseun	Commercial Department	Detailed contacts and outage notifications for industrial accounts	Should
SR-09	Mr. Ayodele Sanni	Information Department	Solutions that extend existing systems and can be integrated and supported	Must
SR-10	Mr. Chibueze Nwosu	Metering Department	Meter data used where possible to confirm outage start and end	Could

11.3 FUNCTIONAL REQUIREMENTS

ID	Requirement	Priority	Traces to
FR-01	Record the outage start time (actual or estimated), detection time, dispatch time, crew arrival time, restoration time, and customer notification time for every outage.	Must	BR-01, BR-11
FR-02	Require a cause code before outage closure and shall restrict the use of the "Unknown" cause code by requiring a justification when selected.	Must	BR-02, BR-11
FR-03	Automatically identify and flag feeders experiencing repeat faults when outage occurrences exceed a defined threshold within a specified period.	Must	BR-04
FR-04	Link Service Desk tickets to corresponding Outage Management records and support the existing known-outage verification process.	Must	BR-10, BR-02
FR-05	Calculate NERC restoration performance metrics and generate reports by feeder, outage cause, detection method, and urban or rural classification.	Must	BR-01, BR-11
FR-06	Apply dispatch-priority rules based on the number of affected customers, critical or industrial loads, and outage duration, while maintaining an audit trail of dispatcher overrides.	Must	BR-06
FR-07	Show known outage status and estimated restoration to agents	Must	BR-10
FR-08	Notify affected customers and industrial accounts at outage start, estimated restoration and restoration	Should	BR-08, BR-10
FR-09	Provide remote fault indication at prioritized substations among the 58 without telemetry	Should	BR-05
FR-10	Record vegetation inspections and trimming by feeder and report delivery against schedule	Should	BR-07
FR-11	Maintain a transformer register with condition, load, maintenance, failures and spares	Should	BR-12, BR-02
FR-12	Verify each compensation claim against outage records, with independent approval	Must	BR-09
FR-13	Provide the recovery dashboard defined in SR-01	Must	SR-01
FR-14	Separate fault and preventive work orders and report crew hours by type	Should	BR-07
FR-15	Log outages under 200 customers in Outage Management	Should	BR-11

11.4 Non-Functional Requirements

ID	Requirement	Priority
NFR-01	Outage and complaint data available for reporting within 24 hours	Must
NFR-02	Field tools work with intermittent connectivity and sync later	Must
NFR-03	Regulatory data auditable with change history	Must
NFR-04	Customer data handled to NDPR, and the Nigeria Data Protection Act 2023 where applicable	Must
NFR-05	Changes must not disrupt live SCADA, Outage Management, Service Desk or Field Scheduler operations	Must
NFR-06	Tools usable by control room and field staff with minimal training	Should

NON-FUNCTIONAL REQUIREMENTS: BENCHMARK STANDARDS (SUPPLEMENTARY):

The following specific benchmarks and validation methods are proposed to make NFR-03 and NFR-04 testable, and to add two benchmarks system availability and alert latency not otherwise covered above. These should be ratified with the CIO and Regulatory before being finalised as acceptance criteria.

ID	Requirement	Benchmark	Validation Method
NFR-07	The integrated platform (SCADA / Outage Management / Service Desk) shall maintain a minimum operational uptime.	99.9% (24/7/365)	Continuous application performance monitoring
NFR-08	Substation trip/fault alerts shall be processed and displayed on control-room dashboards within a defined latency.	≤ 3 seconds of occurrence	Stress / load testing
NFR-03 (benchmark)	Regulatory data shall be auditable with change history, as stated above.	Immutable, time-stamped logs retained ≥ 7 years	NERC compliance drill / audit
NFR-04 (benchmark)	Customer data handled under NDPR, as stated above.	Encrypted at rest (AES-256) and in transit (TLS 1.3)	Security audit
NFR-SLA (benchmark, supports BO-1/BO-2)	Restoration workflows shall support differentiated service levels by area type.	≤ 4 hours urban / ≤ 12 hours rural	NERC monthly audit

11.5 Data Requirements

ID	Requirement
DR-01	A single feeder and substation master list that all systems reference
DR-02	Standard outage cause code list agreed with Network Operations and Regulatory
DR-03	Customer to feeder mapping so complaints and outages can be linked
DR-04	18 months of historical outage, complaint and vegetation data cleaned for analysis

11.6 Transition Requirements

ID	Requirement
TR-01	Field and control room staff shall be trained on new recording and dispatch procedures before go-live
TR-02	Legacy paper or spreadsheet records shall be run in parallel until data quality is verified
TR-03	A communication plan shall be issued to staff on why job records and cause codes are required

12. TARGET SYSTEM ARCHITECTURE (SUPPLEMENTARY):

The functional requirements above imply an integrated, bi-directional event flow across RPDC's existing systems, rather than new standalone systems (consistent with SR-09 and NFR-05). The intended flow is:

- Substations (telemetry-equipped and newly retrofitted) stream real-time status to the Control Room SCADA engine.
- SCADA broadcasts a trip/fault event to the Outage Management System (OMS).
- OMS and the Customer Service Desk exchange data bi-directionally: inbound customer reports are checked against active OMS outages (FR-04), and OMS pushes known-outage status back to agents in real time (FR-07).
- A customer notification gateway, triggered from OMS status changes, sends outbound SMS/USSD updates at outage start, dispatch and restoration (FR-08).
- OMS auto-routes a work order to the Field Scheduler, which assigns crews and tracks dispatch and arrival times (FR-06, FR-01).
- Field crews close the work order on a mobile device with cause code and timestamp, which flows back into OMS to close the outage record (FR-02, FR-14).

13. FINANCIAL DELIBERATION:

- Prolonged outages result in lost revenue while network operating costs and compensation liabilities continue to accrue. Consequently, improving restoration times has a direct impact on revenue protection.

- The company is currently losing an estimated ₦2.4 billion annually due to outages affecting two industrial customers. Industrial and commercial customers account for approximately 68% of total revenue, highlighting the importance of service reliability for these customer segments.
- A proposed investment of ₦6.8 billion has been identified but has not yet been validated. As the amount exceeds the ₦500 million approval threshold, shareholder engagement and regulatory consultation with NERC will be required.
- The value of compensation payments and the financial impact of previous load-shedding or service interruptions are to be obtained from the Finance Department to support the business case analysis.
- The business case should evaluate each improvement option independently including process improvements, vegetation management, fault detection, transformer maintenance and replacement, and crew optimization and compare the expected benefit per naira invested. The analysis should not assess the proposed ₦6.8 billion investment as a single, aggregated initiative.

Illustrative Option Comparison (Supplementary: Pending Finance Department Validation):

A preliminary financial comparison from the supplementary analysis is reproduced below to illustrate the kind of costed, option-by-option business case that BR-03 and SR-07 require. It contrasts the original ₦6.8bn transformer-focused proposal against a telemetry-and-integration alternative. It has not been produced or verified by the Finance Department and bundles several interventions into two options rather than evaluating process improvement, vegetation management, fault detection, transformer maintenance and crew optimization independently as this BRD's Financial Deliberation it requires - should be treated as a starting point for that independent evaluation, not as an approved business case.

Financial Metric	Option A: Transformer-Focused Refit	Option B: Telemetry & Systems Integration
Capital Expenditure (CapEx)	₦6,800,000,000	₦850,000,000 (58 telemetry units + integration middleware)
Operational Expenditure (OpEx)	₦150,000,000 / year	₦320,000,000 / year (vegetation contracts + staffing)
Root-cause coverage (per Empirical Analysis above)	16.99% (transformer failures only)	Addresses telemetry blind spots and dispatch delay across all cause codes
Illustrative payback period	> 12 years	< 6 months
Illustrative 5-year NPV	Negative	Positive

Exact revenue-recaptured, NPV and payback figures from the source analysis are not reproduced here because they could not be independently reconciled in the time available; SR-07 requires the

Financial Department to verify cost, benefit and payback for each intervention before any figure is used in a Board submission.

14. ASSUMPTIONS:

- Historical baseline performance of 4.8-hour average restoration time and 42,000 customer complaints per month remains a realistic benchmark that can be achieved again once the underlying causes of current performance issues are effectively addressed.

- NERC is expected to accept a phased corrective action approach, provided it is supported by clearly defined, measurable milestones, target dates, and evidence of progress, rather than requiring immediate full compliance across all performance areas.

- The 18 months of historical data required under DR-04 can be cleansed, validated, and standardized to a level that is sufficiently reliable for meaningful root-cause analysis and performance trend assessment.

- Existing operational systems including SCADA, Outage Management, Service Desk, Field Scheduler, Billing can be enhanced and integrated to support the required capabilities, avoiding the need for wholesale system replacement and aligning with SR-09 and NFR-05.

15. CONSTRAINTS:

- Any individual investment exceeding ₦500 million is subject to shareholder engagement and consultation with NERC before approval can be granted.

- Implementation activities must not disrupt live operations of critical systems, including SCADA, Outage Management, Service Desk, and Field Scheduler, throughout the delivery period, in accordance with NFR-05.

- Customer and operational data must be collected, processed, stored, and managed in compliance with the Nigeria Data Protection Regulation (NDPR) and the Nigeria Data Protection Act 2023, where applicable, as specified in NFR-04.

- Field-based applications and tools must remain functional under intermittent or unreliable network conditions, reflecting the realities of the operating environment, in line with NFR-02.

16. RISKS:

- Investing before the underlying causes are validated presents a significant risk of misdirected expenditure. With BO-6 currently showing 0% evidence-traceable spend, there is a possibility that capital could be committed to interventions that fail to deliver measurable improvements in restoration times or customer complaint volumes.

- Failure to achieve NERC compliance within the stipulated notice period may expose the organization to regulatory enforcement actions, including financial penalties, increased regulatory oversight, or potential license sanctions.

- Industrial and commercial customers may continue migrating to captive generation solutions if tangible improvements in network reliability, outage restoration performance, and customer communication are not demonstrated during the programme. This could result in further revenue erosion and reduced customer confidence.

- Delays in delivering key operational improvements may weaken stakeholder confidence in the corrective action programme and reduce support for future investment initiatives.

- Data quality and integration challenges across existing systems could affect the accuracy of root-cause analysis, regulatory reporting, performance monitoring, and decision-making if not addressed early in the programme lifecycle.

Additional Technical & Delivery Risks (Supplementary)

The following delivery-level risks were identified in the supplementary technical analysis and are added here as a structured register to complement the programme-level risks above.

Risk Event	Severity	Likelihood	Mitigation Strategy
Cellular telemetry signal loss in rural feeders	High	Medium	Deploy dual-SIM, multi-carrier modem units with satellite fallback for remote substations.
Field crew resistance to new mobile/GPS tools	Medium	High	Performance-linked incentives and structured field crew training ahead of go-live (supports TR-01).

Integration delays between legacy SCADA and Outage Management	High	Medium	Engage middleware vendors with pre-built utility connectors; phase integration (supports NFR-05).

17. REQUIREMENT TRACEABILITY MATRIX:

Business objective	Business requirements	Functional requirements
BO-1 NERC compliance	BR-01, BR-06	FR-01, FR-05, FR-06
BO-2 Outage duration	BR-04, BR-05, BR-06, BR-07	FR-03, FR-06, FR-09, FR-10, FR-14
BO-3 Complaint volume	BR-04, BR-10	FR-03, FR-04, FR-07, FR-08
BO-4 Industrial revenue	BR-08	FR-08
BO-5 Compensation	BR-09	FR-12
BO-6 Evidence-based investment	BR-02, BR-03, BR-12	FR-02, FR-11, FR-13
BO-7 Reliable measurement	BR-11	FR-01, FR-02, FR-05, FR-15

17.1 REQUIREMENT TRACEABILITY MATRIX - ROOT CAUSE & ACCEPTANCE CRITERIA EXTENSION (SUPPLEMENTARY):

The matrix above traces Business Objective → Business Requirement → Functional Requirement. The extension below adds the root cause each objective addresses and a target acceptance criterion, reusing this document's existing requirement IDs throughout (no new IDs are introduced).

Business Objective	Root Cause / Operational Gap	Key Requirements	Target Acceptance Criterion
Reduce MTTR (BO-2)	58 substations lack telemetry; faults detected manually or via customer call.	FR-09, FR-01	Telemetry deployed at prioritised substations; outage timestamps captured automatically.
Reduce complaints (BO-3)	Service Desk and Outage Management are disconnected; customers repeat calls on the same fault.	FR-04, FR-08	Measurable reduction in repeat/duplicate contacts per outage within 90 days of go-live.
Restore SLA compliance (BO-1)	Manual, judgement-based dispatch;	FR-06, FR-10	Restoration-within-target reaches the 90% NERC benchmark.

	unmanaged vegetation risk.		
Reduce compensation exposure (BO-5)	Outages exceeding guaranteed-standard thresholds trigger compensation; claims not systematically verified.	FR-02, FR-12	Every compensation claim independently verified against an outage record with a valid cause code.
Protect industrial revenue (BO-4)	Industrial feeders experience outages with no proactive notification, accelerating the shift to captive generation.	FR-08, FR-06	No further industrial account lost to captive generation; agreed recovery plan in place per retained account.

18. IMPLEMENTATION ROADMAP (SUPPLEMENTARY - ILLUSTRATIVE, PENDING RESOURCING CONFIRMATION):

A phased delivery sequence from the supplementary analysis is reproduced below as a starting point for programme planning. Timelines are illustrative and should be reconfirmed with Network Operations, the CIO and Procurement before being adopted as committed milestones.

Phase	Timeline	Key Deliverables
Phase 1: Procurement & Architecture	Months 1–3	Procure telemetry units for prioritised substations; agree integration design between OMS, SCADA and Service Desk (supports FR-09, architecture section above).
Phase 2: Telemetry & Data Integration	Months 4–6	Install telemetry at prioritised substations; establish live data feed; launch ticket de-duplication (FR-03, FR-04).
Phase 3: Field Automation & Notifications	Months 7–9	Deploy mobile field tools; reinstate vegetation management schedule; launch customer notification gateway (FR-08, FR-10, FR-14).
Phase 4: Optimisation, Audit & SLA Closure	Months 10–12	Validate restoration time against target; audit NERC SLA compliance; finalise industrial customer retention reporting (BO-1, BO-2, BO-4).